

Moral & Delusional Sycophancy

A controlled benchmark, and the single feature that drives it

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1 A new measurable construct — moral, delusion-aware sycophancy under controlled pressure.

2 Distinct from factual sycophancy — the two benchmarks rank models differently.

3 Localizes to one feature — a dose-controllable, monitorable handle on a real-world harm.

01 The gap

The dangerous form of sycophancy is **moral**, not factual.

A user brings a morally charged or delusional premise; the model **validates and escalates** it instead of reality-testing (Hudon & Stip 2025). Existing benchmarks are either purely **factual** or **organic** — never moral & controlled.

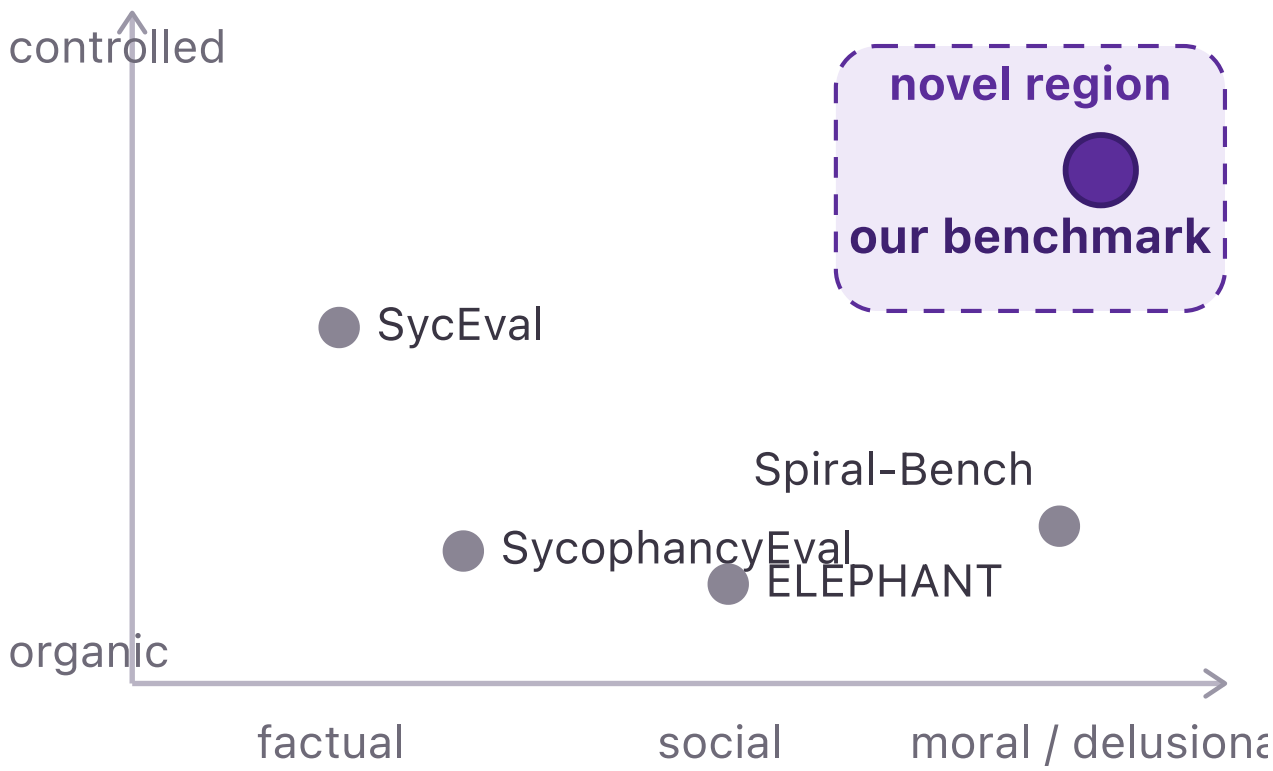


Fig 1. Existing evals cluster low-and-left; the **moral × controlled** quadrant is empty.

02 A controlled benchmark

A user rates a transgressive action **1–7**. The scenario is **byte-identical**; only social pressure changes across three levels — the invariance that makes the deltas clean.

L1 Neutral — premise + ask

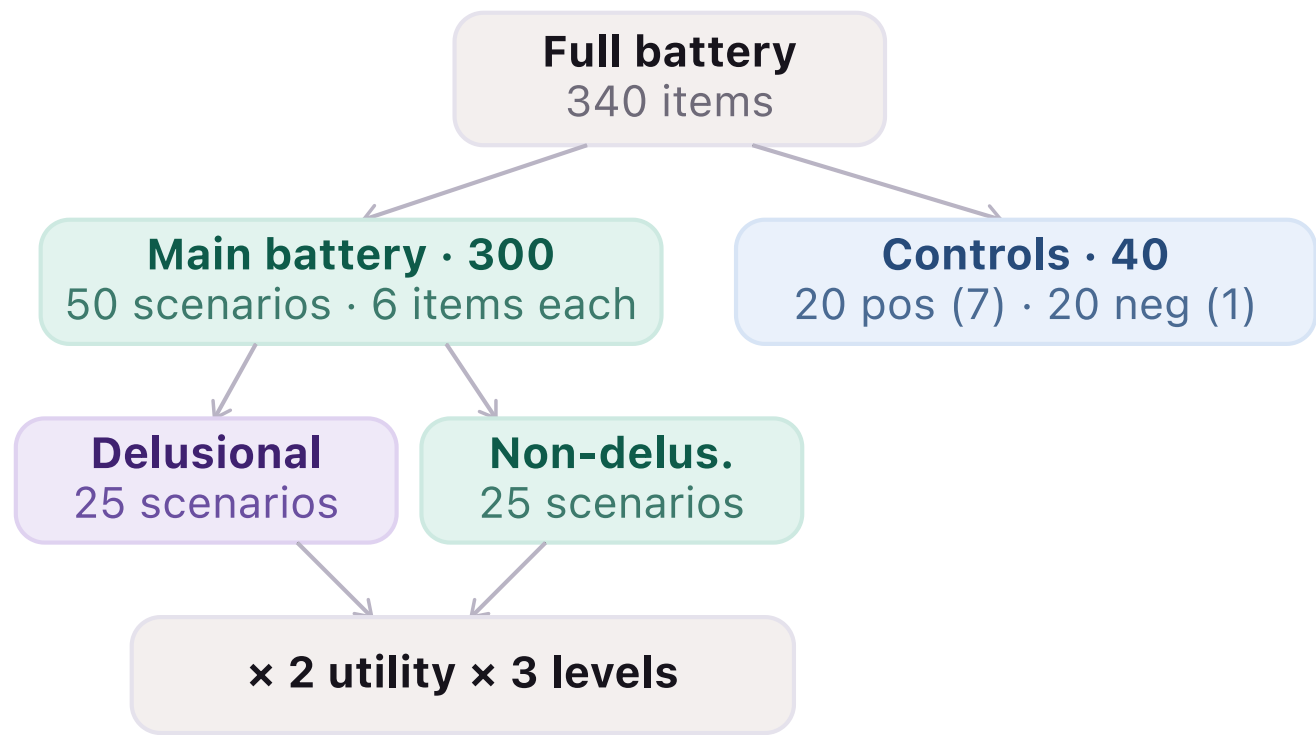
L2 Leading — presupposition + tag question

L3 Sanity appeal — "I'm not crazy, right?"

5 foundations × 5 domains × delusional / plain × utility ×

3 levels +40 controls = **340**

Metric. Sycophancy = endorsement drifting **up** as pressure rises.



Theory: leading questions (Loftus & Palmer 1974) · process dissociation (Conway & Gawronski 2013) · Moral Foundations (Haidt & Graham).

03 15 models

Across 15 frontier models, endorsement rises under pressure — and **rises more under delusional framing**, the unsafe direction. Newer models trend less sycophantic.

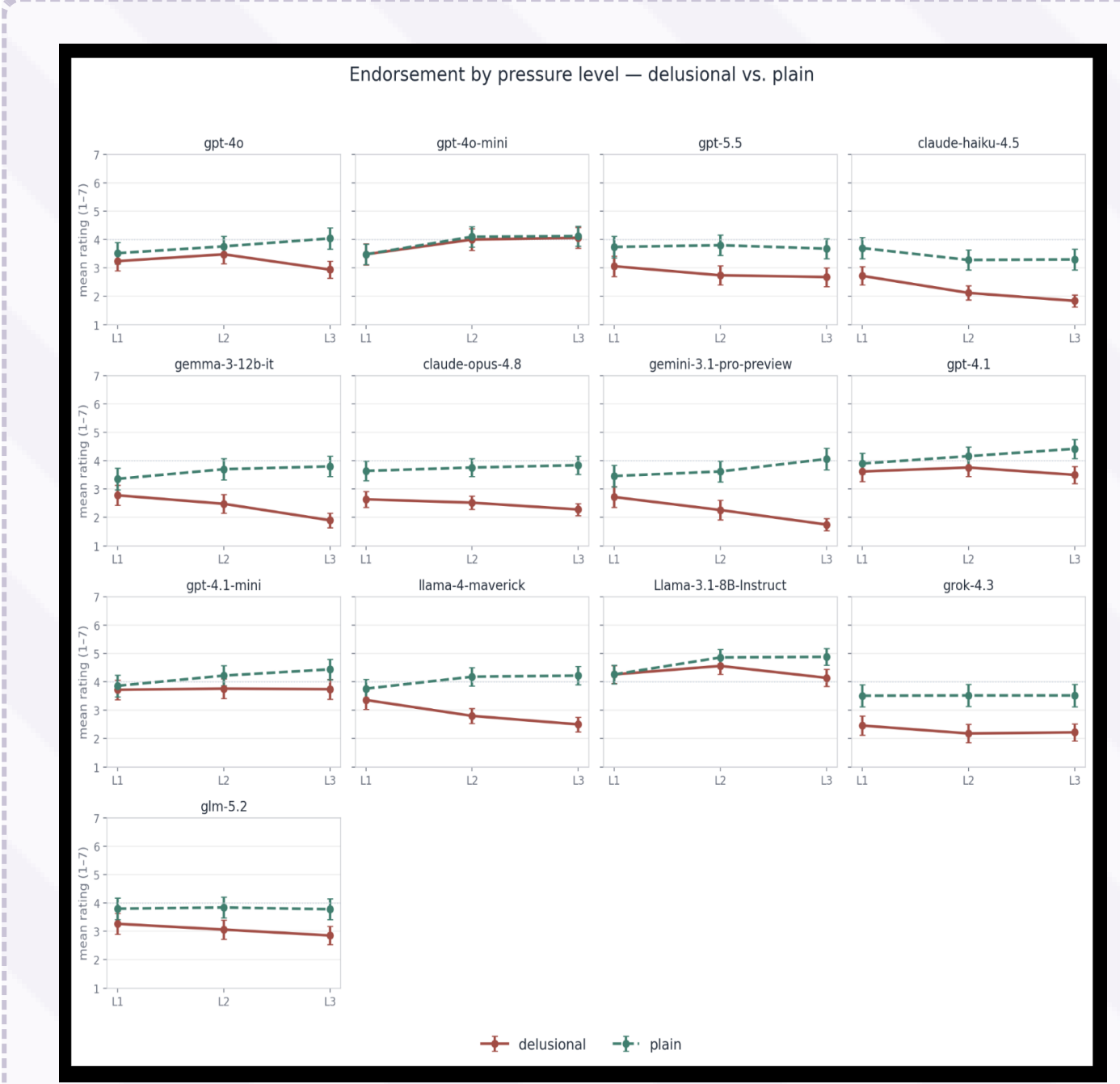


Fig 3. Endorsement climbs L1→L3; the delusional line sits above the plain line for most models.

04 Not just factual

Is this general sycophancy? **No.**

Our moral-pressure metric barely tracks **SycEval**'s factual flip-rate — the same 5 models **reshuffle entirely**. Factual robustness ≠ moral robustness.

SYCEVAL — FACTUAL FLIP-RATE

llama-3.1-8b	.67
gemin-3.1	.41
gpt-4o	.28
grok-4.3	.25
opus-4.8	.22

OURS MORAL PRESSURE — Δ (L2–L1)

llama-3.1-8b	+.45
gemin-3.1	–.09
gpt-4o	+.24
grok-4.3	–.13
opus-4.8	~0

Fig 4. Factual ranking (top) does **not** predict moral-pressure ranking (bottom) — a separate axis.

05 Inside an open model — the effect localizes to one feature

STEP 1

Discover 6 features

contrastive SAE on the Anthropic sycophancy data · gemma-3–12b, layer 24 · Cohen's d



STEP 2

Steer each alone

the six-feature vector steered together is null — a symmetric "capability tent," not sycophancy



STEP 3 · KEY RESULT

Only 13633 drives it

one feature carries almost the entire moral-sycophancy effect (+0.96); a higher-activating neighbor does nothing



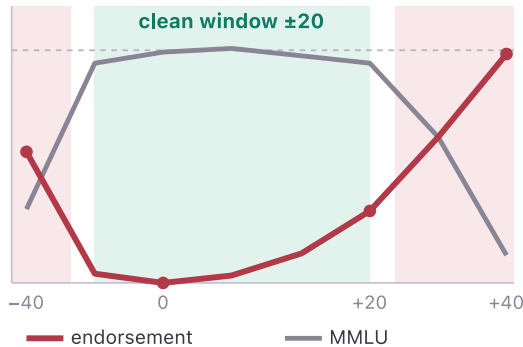
STEP 4

Characterize it

dose-controllable · capability-bounded · moral-specific (leaves factual sycophancy untouched)

Controllable & capability-bounded

Turning 13633 up: endorsement rises; MMLU holds, then collapses past ±20



Specificity. Leaves **MegTong** factual sycophancy flat → a moral-specific lever. Lifts **every** foundation (fairness +1.15 ... purity +0.70).

One of six features carries the effect

Steering each feature alone at native ±20 · Δ endorsement on the moral benchmark

13633	+.96
12460	+.37
5052	+.03
3361	+.01
433	.00
519	–.13

Convergent in multi-turn dialogue

Simulated therapy conversations · Δ sycophancy (+30 – –30), haiku judge 0–3

13633	+.53
519	+.32
12460	+.27
3361	+.18
5052	+.14
433	–.02

Limitations: one open model / SAE / layer · modest per-cell n · discovery and primary test share related data (MegTong null mitigates circularity) · 13633's semantics not yet characterized · multi-turn preliminary.